

Homework 1

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Math 38

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Question 1. (10 points) Recall the connected relation of vertices - *given a graph G and vertices $u, v \in G$, u and v are connected if there is a uv -path in G .* Check if this is an equivalence relation and justify your answer.

Proof. Firstly, check if connectivity is reflexive. For $u \in V(G)$ in a graph $G(V, E)$, it is trivial that the vertices u and u are connected because the path is simply a sequence of length zero. Next, check if connectivity is symmetric. If u and v are connected, then there exists a path uv -path in G , which composes of a sequence $u \dots v$. By simply reversing the ordering of the sequence to $v \dots u$, this implies that there exists a path vu in G . Therefore, v and u are connected.

Lastly, check if connectivity is transitive. Suppose vertices u and v are connected, and let us define a new vertex $w \in G$ such that v and w are connected. This then implies that there is a path from u to v and a path from v to w . By combining these two paths, we can create a new composite walk, one starting from u and ending to w . By removing the redundant sequences, this can be reduced to a path uw . Therefore, u and w are connected.

The three requirements are satisfied, and connectivity is therefore a equivalence relation. $\square$

Question 2. (10 points) Prove or disprove: The complement of a simple disconnected graph is connected.

Proof. Suppose we have a simple disconnected graph G . Now, suppose we pick two arbitrary vertices $u, v \in V$. There are only two cases we analyze. If u and v are *not adjacent* in G , in $\overline{G}$, the two vertices will be adjacent and thus connected because of the existence of a path uv .

Now, suppose u and v are *adjacent* in G . By definition of a disconnected graph, there must exist a vertex $w \in V$ such that w is in a different component than $\{u, v\}$. Therefore, in $\overline{G}$, there exists paths uw and wv . Since these two paths exist, by transitivity, there implies a path uv .

Therefore, for all pairs of arbitrary vertices u and v , uv is connected in $\overline{G}$, and $\overline{G}$ is connected.

□

Question 3. (10 points) Prove or disprove:

- (a) Every Eulerian bipartite graph has an even number of edges.
- (b) Every Eulerian simple graph with an even number of vertices has an even number of edges.

Proof. For part (a), we define a graph $G(A \cup B, E)$. Since the graph is bipartite, every edge goes from A to B . Therefore, the total number of edges can be found by adding the degrees of each vertex across one independent set. However, we know that the graph is also Eulerian, forcing every vertex to have an even degree. Since the sum of even numbers is also even, the graph has an even number of edges.

For part (b), let G be a simple Eulerian graph with k vertices, where k is even. By the handshake formula,

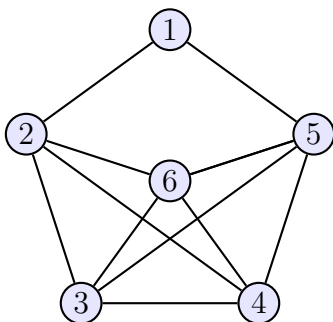
$$2|E| = \sum_{v_i \in V}^k d_G(v_i).$$

Since G is Eulerian, every vertex has an even degree. For each vertex, where x_i represents an integer,

$$2|E| = \sum_{i=1}^k 2x_i \implies |E| = \sum_{i=1}^k x_i,$$

so the total number of edges is a sum of k positive integers. However, there is no guaranty that the sum of k positive integers is even. Therefore, we disprove that $|E|$ is even. We can further disprove this claim by counterexample, as shown below:

Graph G



The graph G has six vertices and is Eulerian because each vertex has an even degree. However, there are a total of 11 edges, which is an odd number. $\square$

Question 4. (10 points) Let v be a cut-vertex of a simple graph G . Prove that $\overline{G} - v$ is connected.

Proof. This is similar to problem two, so let us follow a similar approach. Let v be a cut-vertex of a simple graph G . Based on the definition of cut-vertices, the removal of v results in at least two components. To prove that $\overline{G} - v$ is connected, let us define two arbitrary vertices x and y in $V(G) - \{v\}$.

We consider two cases. If x and y are in different components in $G - v$, there is no edge between them in G . Thus, in $\overline{G} - v$, there exists an edge xy , and x and y are connected.

If, however, x and y are in the same component in $G - \{v\}$, since v is a cut-vertex, there exists at least one other component in $G - \{v\}$. We define a new vertex w such that w is in a different component from x and y . Because w is in a different component, there are no edges xw or wy in G . Consequently, in $\overline{G} - \{v\}$, there exists an edge xw and an edge wy . Therefore, there exists a path between x and y by using w as an inner vertex. Similar reasoning works for a cut-vertex that forms more than two components.

Since a path exists between any arbitrary vertices x and y in all cases, $\overline{G} - \{v\}$ is connected. $\square$